

MyFirstgame

A fast paced pvp shooter

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Component Overview

Frame

Client

GameStore

Renderer

Client-Specific Networking

Client-Specific GameStateUpdater

Frame

Used by both

StageManager

GameUpdate

Base Networking

Frame

Server

Server-Specific Networking

Server-Specific GameStateUpdater

Dependencies

Frame

Dependencies

SFML::Window

SFML::Graphics

SFML::Network

SFML::Audio

nlohman/json

Plog

Frame

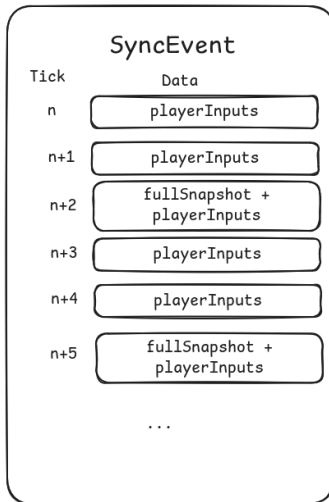
Build System

Cmake 3.28+

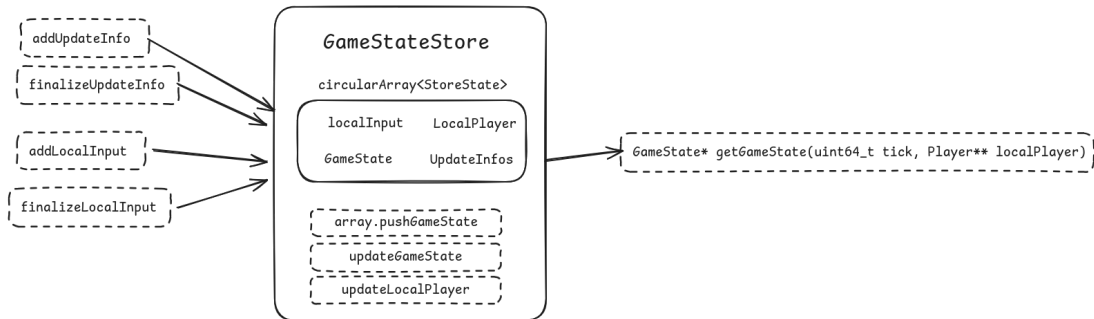
c++17

CI is testing
build on linux

Code Insight: GameStateStore



Code Insight: GameStateStore



Code Insight: GameUpdates

```
class GameStateUpdater
{
private:
public:
    GameStateUpdater(){};
    ~GameStateUpdater(){};
    virtual void setPlayerVelocity(Player& player, sf::Vector2f velocity) = 0;
    virtual void deltaMovePlayer(Player& player, sf::Vector2f deltaPosition) = 0;
    virtual void absoluteMovePlayer(Player& player, sf::Vector2f absolutePosition) = 0;
    virtual void setPlayerOnGround(Player& player, bool isOnGround) = 0;
    virtual void setPlayerProjectileCooldown(Player& player, int16_t cooldown) = 0;
    virtual void setPlayerFacing(Player& player, bool facing) = 0;
    virtual void addProjectile(Projectile& projectile) = 0;
    virtual void setProjectileIds(OBJECT_ID_TYPE id) = 0;
    virtual void addPlayer(Player& player) = 0;
    virtual void projectileHitPlayer(Projectile& projectile, Player& player) = 0;
    virtual void setReadyToPlay(Player& player, bool ready) = 0;
    virtual void registerRemoveProjectile(Projectile& projectile) = 0;
    virtual void removeRegisteredProjectiles() = 0;
    virtual void registerRemovePlayer(Player& projectile) = 0;
    virtual void removeRegisteredPlayers() = 0;
};
```

Code Insight: GameUpdates

```
virtual void projectileHitPlayer(Projectile& projectile, Player& player) override {};
```

```
virtual void projectileHitPlayer(Projectile& projectile, Player& player) override {gameState.getPlayer(player.getId()).setHealth(0);};
```

Lessons learned

Planning before implementing

First Multiplayer is hard

Debugging is useful

Live Demo



Thank you for your attention!

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